

Adjustable Robust Optimization

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CR03 - Robust combinatorial optimization, ENS-Lyon

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Little reminder

- We talked about "static" robust optimization problems:

$$\begin{aligned}
 \min \quad & \mathbf{c}^\top \mathbf{x} \\
 \text{s.t.} \quad & \mathbf{a}_i(\boldsymbol{\xi})^\top \mathbf{x} \geq b_i \\
 & \mathbf{x} \in \mathcal{X}
 \end{aligned}
 \quad \forall \boldsymbol{\xi} \in \Xi_i, \forall i \in [m]$$

non-empty & bounded

- "Static" refers to all decisions being taken before the realization of uncertainty ("here-and-now").
- Under this assumption many robust optimization problems may be reformulated as deterministic-equivalent problems.
- The reformulation often preserves the class of complexity of the deterministic problem.

Little reminder

$$\begin{array}{ll}
 \min & \mathbf{c}^\top \mathbf{x} \\
 \text{s.t.} & \mathbf{a}_i(\boldsymbol{\xi})^\top \mathbf{x} \geq b_i \\
 & \mathbf{x} \in \mathcal{X}
 \end{array}
 \quad \forall \boldsymbol{\xi} \in \Xi_i, \forall i \in [m]$$

$\hookrightarrow \boldsymbol{\xi} \in \mathbb{R}^{n_i} \mid D_i \boldsymbol{\xi} \geq d_i$

Write $\min_{\boldsymbol{\xi} \in \Xi_i} \mathbf{a}_i(\boldsymbol{\xi})^\top \mathbf{x} \geq b_i \quad \forall i \in [m]$ with $\mathbf{a}_i(\boldsymbol{\xi}) = A_i \boldsymbol{\xi}$ then

$$\begin{array}{ll}
 \min_{\substack{\mathbf{x} \in \mathcal{X} \\ \boldsymbol{\xi} \in \Xi_i}} & \mathbf{x}^\top A_i \boldsymbol{\xi} \\
 \text{s.t.} &
 \end{array}
 = \max_{\substack{\mathbf{u} \geq 0 \\ D_i^\top \mathbf{u} = \mathbf{x}^\top A_i}} d_i^\top \mathbf{u} \geq b_i$$

drop the max

Reform as a
deterministic
problem

$$\begin{array}{ll}
 \min & \mathbf{c}^\top \mathbf{x} \\
 \text{s.t.} & \mathbf{a}_i(\boldsymbol{\xi})^\top \mathbf{x} \geq b_i \\
 & \mathbf{x} \in \mathcal{X}
 \end{array}
 \quad \forall \boldsymbol{\xi} \in \Xi_i, \forall i \in [m]$$

$\hookrightarrow \boldsymbol{\xi} \in \mathbb{R}^{n_i} \mid D_i \boldsymbol{\xi} \geq d_i$

$$\longleftrightarrow
 \begin{array}{ll}
 \min_{\mathbf{x} \in \mathcal{X}} & \mathbf{c}^\top \mathbf{x} \\
 \text{s.t.} & \left. \begin{array}{l} d_i^\top \mathbf{u}^i \geq b_i \\ D_i^\top \mathbf{u}^i = \mathbf{x}^\top A_i \\ \mathbf{u}^i \geq 0 \end{array} \right\} \forall i \in [m]
 \end{array}$$

Today: Adjustable robust optimization

Static model :

Decide x $\rightarrow t$

Two-stage model:

Decide x
(Here-and-now)
(First stage)

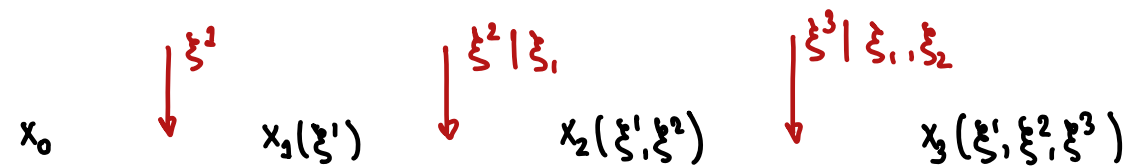
$\xi \in \Xi$
revealed

Adjust with $y(x, \xi)$
(Wait-and-see)
(Recourse)
(Second stage) $\rightarrow t$

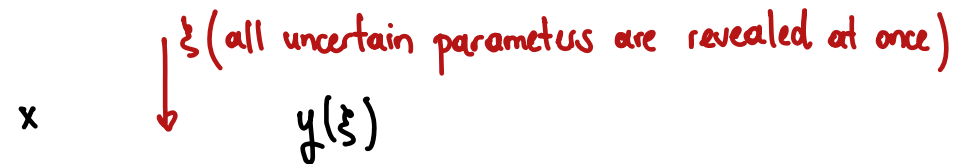
- The variables that can "adjust/adapt" to the realization of uncertainty are called "adjustable/adaptable" or "recourse" variables.

Today: Adjustable robust optimization

Sequential decision-making under uncertainty (multi-stage)



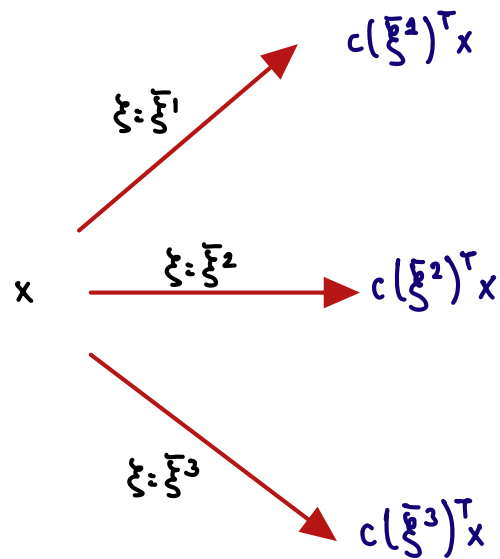
In the two-stage setting :



Today: Adjustable robust optimization

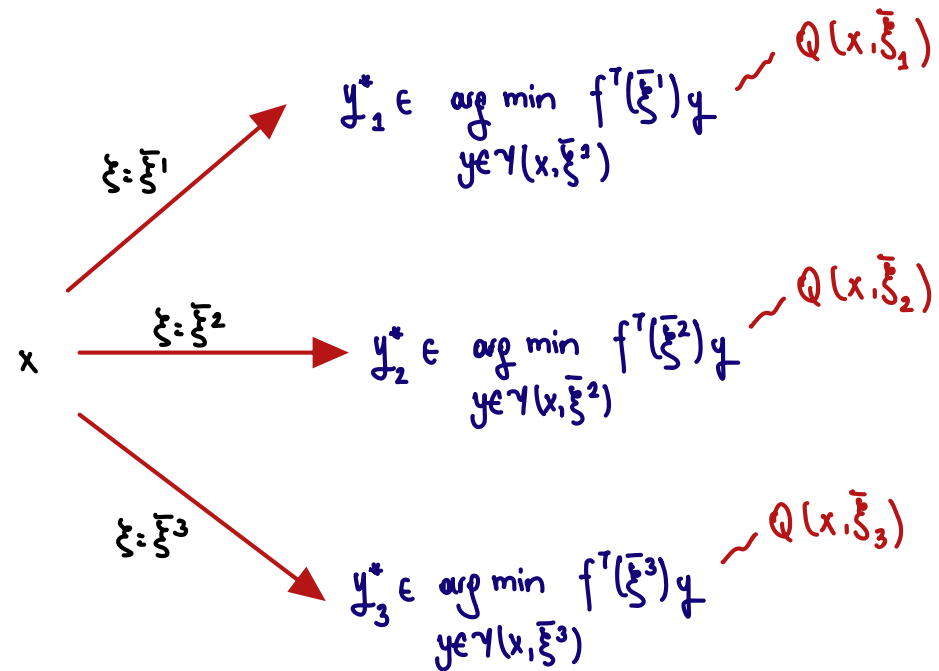
If we think about it with discrete uncertainty realisations say $\Xi = \{\bar{\xi}^1, \bar{\xi}^2, \bar{\xi}^3\}$

Static



$$\min_{x \in X} \max_{\xi \in \Xi} c(\xi)^T x$$

Two-stage / Adjustable



Today: Adjustable robust optimization

How do we compare two solutions x^1 & x^2 ?

we would like to optimize some function $\rho_{\xi \in \Xi} (Q(x, \xi))$

Expectation $\mathbb{E}_{\xi \in \Xi}^{\mathbb{P}} [Q(x, \xi)]$

CVaR $\text{CVaR}_{\alpha, \xi \in \Xi}^{\mathbb{P}} (Q(x, \xi)) = \inf_{t \in \mathbb{R}} t + \frac{1}{1-\alpha} \mathbb{E} [Q(x, \xi) - t]_+$

→ Max (Robust)

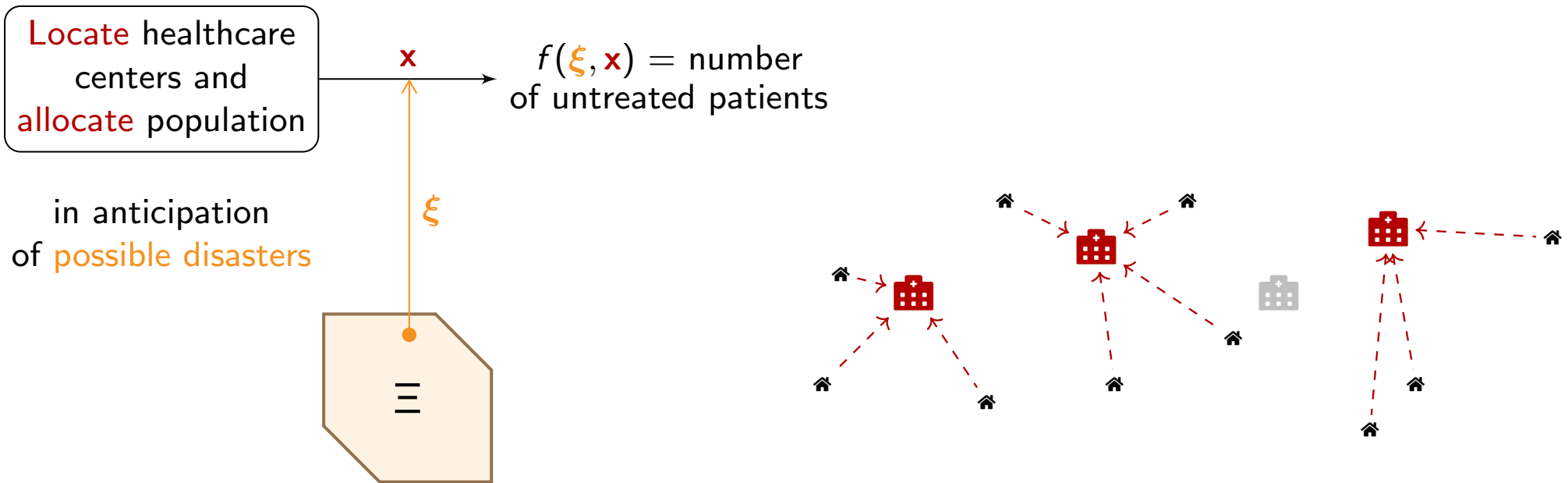
$\max_{\xi \in \Xi} Q(x, \xi)$

Max $\mathbb{E}$ over a family of dist.

$\max_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\xi \in \Xi}^{\mathbb{P}} [Q(x, \xi)] \sim \text{distributionally robust}$

robust $\left\{ \begin{array}{l} \min_{x \in X} c^T x + \rho_{\xi \in \Xi} (Q(x, \xi)) \\ \min_{x \in X} c^T x + \max_{\xi \in \Xi} Q(x, \xi) \end{array} \right\}$ we are often only interested in x^* as output of this model

Today: Adjustable robust optimization



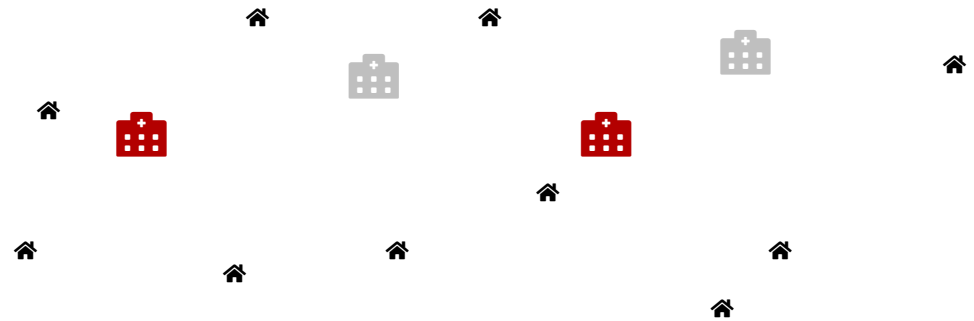
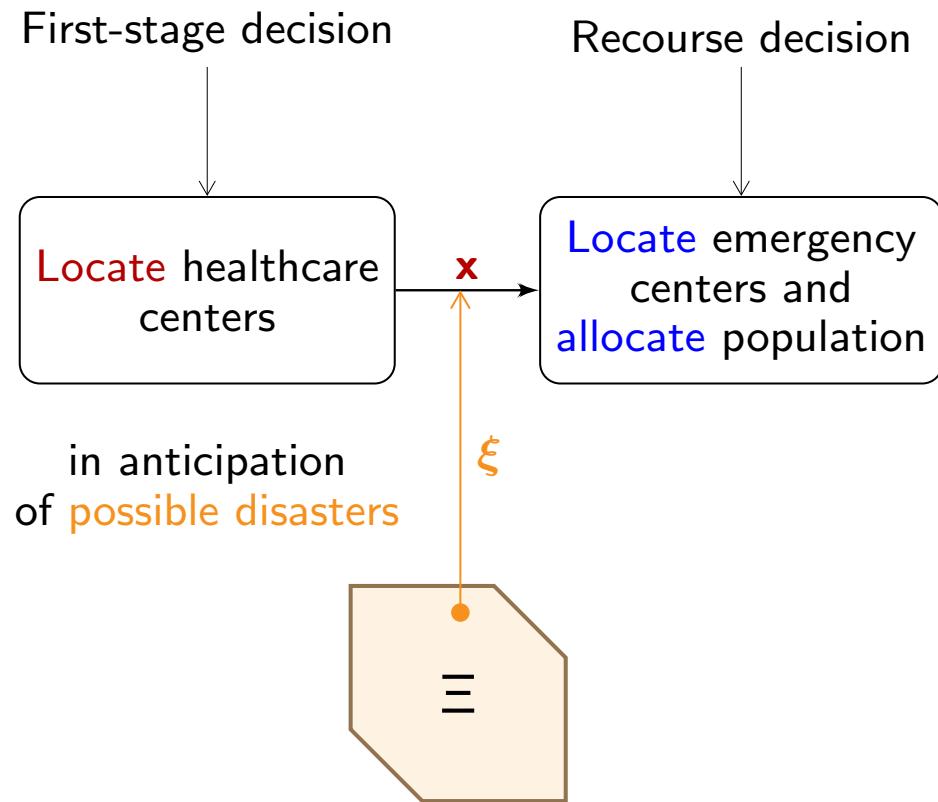
Static model

$$\min_{\mathbf{x} \in \mathcal{X}} \max_{\xi \in \Xi} f(\xi, \mathbf{x})$$

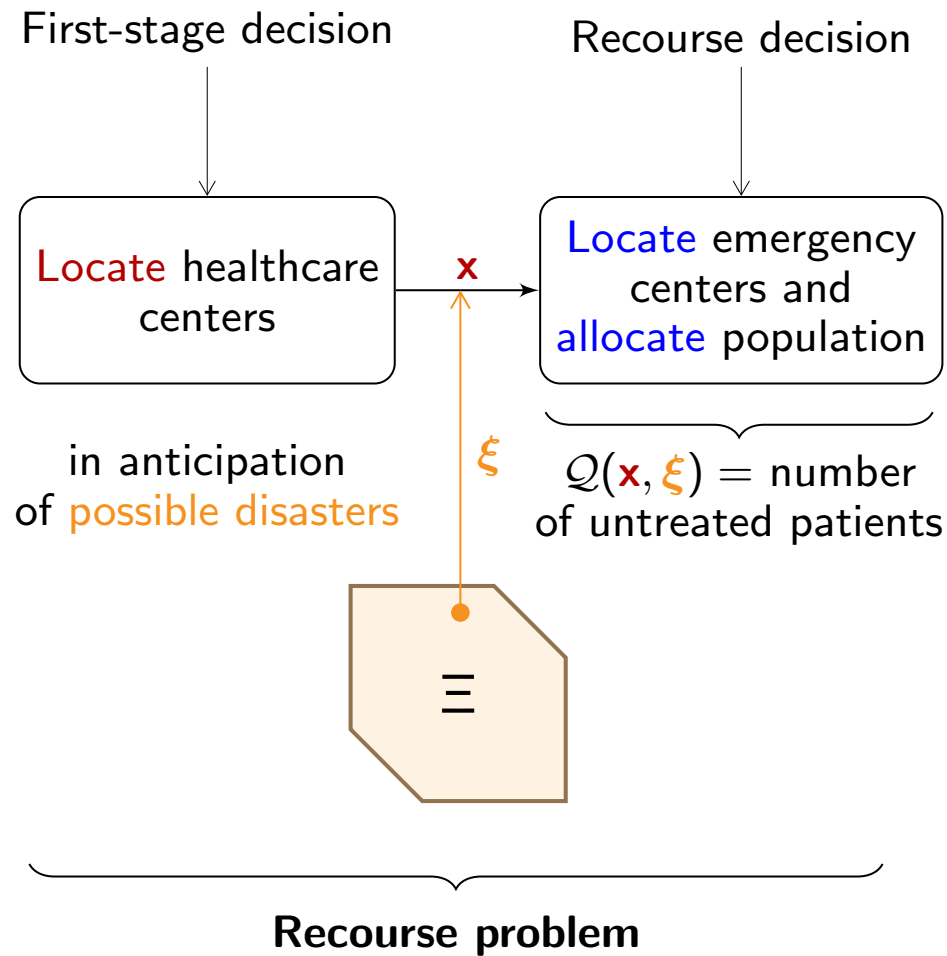
Any $\mathbf{x}$ should be feasible for all considered disaster scenarios

Allocation decisions should be made in advance of the disaster.

Today: Adjustable robust optimization

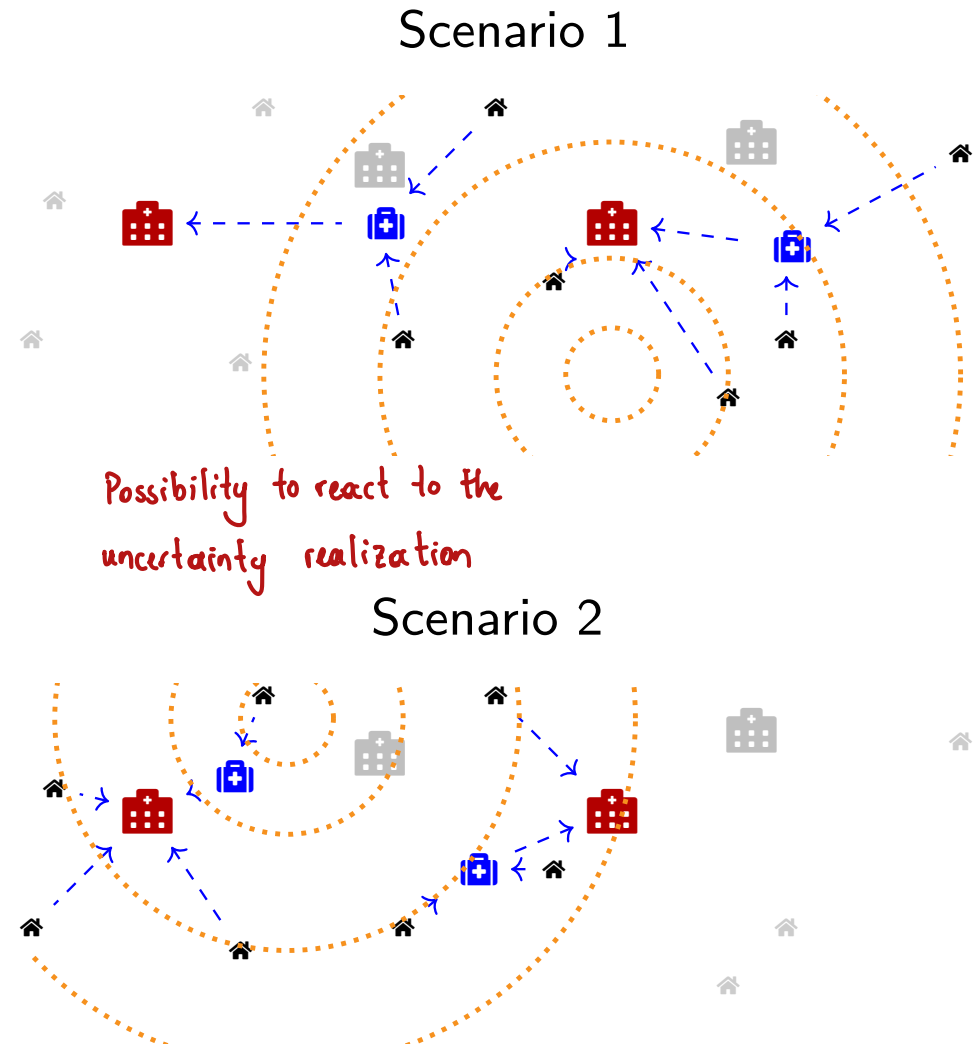


Today: Adjustable robust optimization

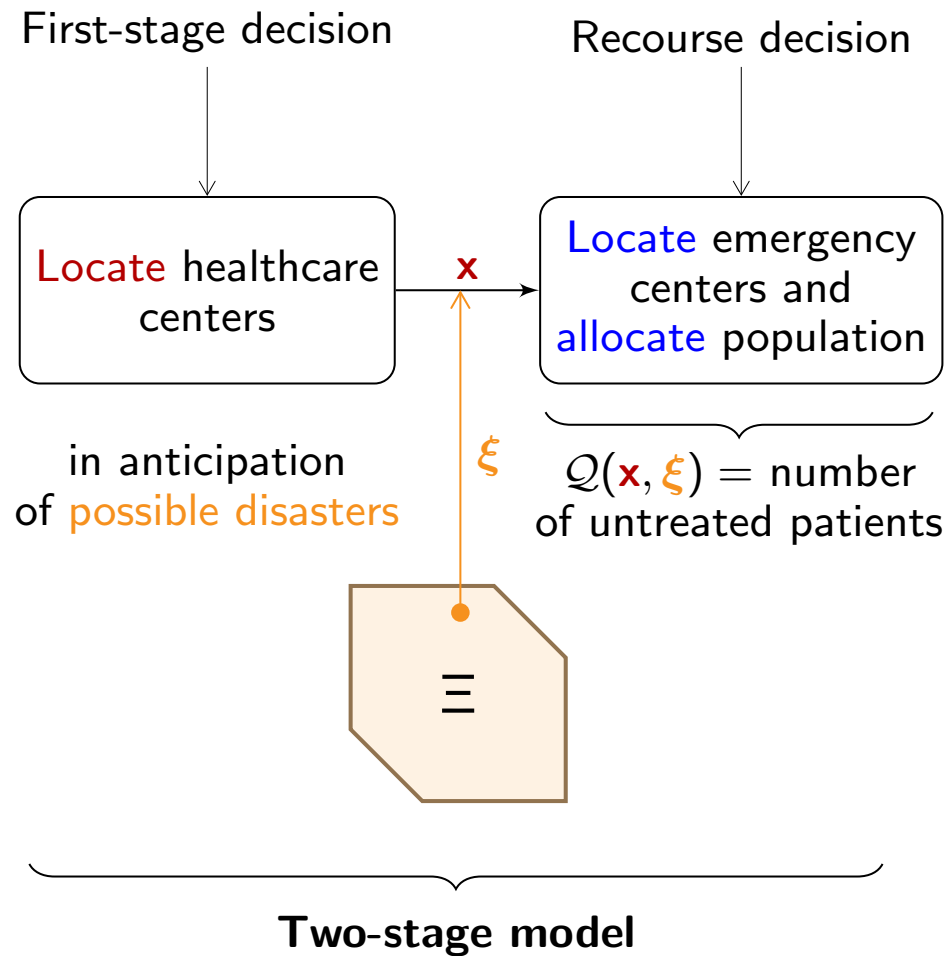


$$Q(\mathbf{x}, \xi) = \min \{f(\xi, \mathbf{x}, \mathbf{y}) \mid \mathbf{y} \in \mathcal{Y}(\mathbf{x}, \xi)\}$$

given a location decision
& a disaster realization



Today: Adjustable robust optimization



$$\min_{\mathbf{x} \in \mathcal{X}} \max_{\xi \in \Xi} Q(\mathbf{x}, \xi)$$

~ Decide $\mathbf{x}$ taking uncertainty realization & the optimal recourse into account

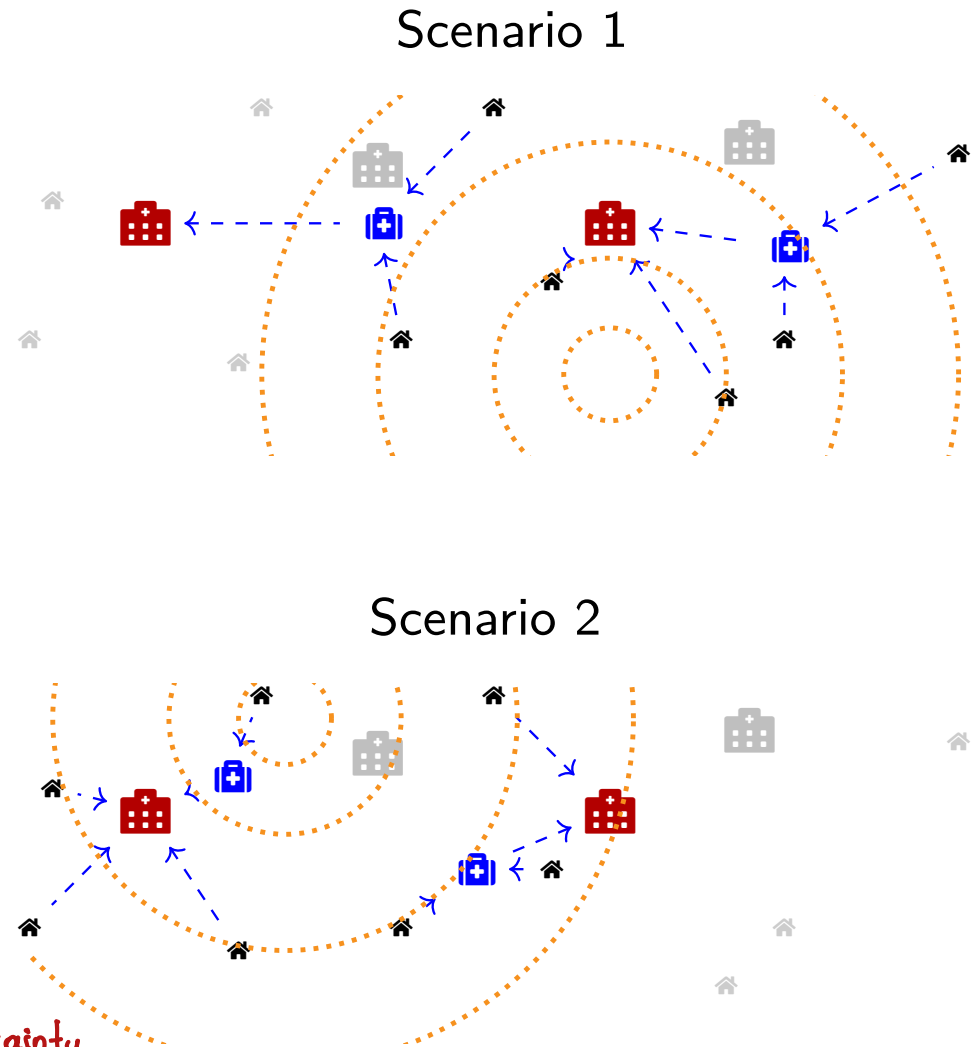


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Two-stage RO: Problem definition

- $\mathcal{X} \subseteq \mathbb{R}^{n_x^c} \times \mathbb{Z}^{n_x^d}$: first-stage feasible region
- ξ : uncertain vector with support $\Xi \subseteq \mathbb{R}^{n_\xi}$
- $\Xi \subseteq \mathbb{R}^{n_\xi}$: uncertainty set, non-empty and compact
- $\mathcal{Y} \subseteq \mathbb{R}^{n_y^c} \times \mathbb{Z}^{n_y^d}$: recourse feasible region

$$\min_{\mathbf{x} \in \mathcal{X}} \quad \mathbf{c}^\top \mathbf{x} + \max_{\xi \in \Xi} Q(\mathbf{x}, \xi) \quad (2\text{ARO})$$

with

$$\begin{aligned} Q(\mathbf{x}, \xi) = \min \quad & \mathbf{f}(\xi)^\top \mathbf{y} \\ \text{s.t.} \quad & \mathbf{W}(\xi) \mathbf{y} \leq \mathbf{h}(\xi) - \mathbf{T}(\xi) \mathbf{x} \\ & \mathbf{y} \in \mathcal{Y} \end{aligned}$$

linking constraints

Remark

In writing (2ARO) we assume that $Q(\mathbf{x}, \xi)$ is an upper semi-continuous function in $\xi \in \Xi$.

Otherwise we should write

$$\min_{\mathbf{x} \in \mathcal{X}} \quad \mathbf{c}^\top \mathbf{x} + \sup_{\xi \in \Xi} Q(\mathbf{x}, \xi)$$

Two-stage RO: Problem definition

Hypothesis and Notation

- $\mathcal{X}$ and $\mathcal{Y}$ are linearly constrained
- Ξ is polyhedral, *i.e.*, $\Xi = \{\boldsymbol{\xi} \in \mathbb{R}_+^{n_\xi} \mid \mathbf{D}\boldsymbol{\xi} \leq \mathbf{d}\}$
- Data is affine in $\boldsymbol{\xi}$:

can have
discrete or continuous
decision variables

$$\mathbf{h}(\boldsymbol{\xi}) = \begin{bmatrix} 1 + \xi_1 \\ 1 - \xi_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ \xi_1 \\ \xi_2 \end{bmatrix} = \mathbf{H}\tilde{\boldsymbol{\xi}}$$

$$\begin{aligned} \mathbf{T}(\boldsymbol{\xi}) &= \begin{bmatrix} 2 + \xi_1 - \xi_2 & 1 - \xi_1 + \xi_2 \\ 3 - \xi_2 & 2 - \xi_1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix} \xi_1 + \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} \xi_2 \\ &= \mathbf{T}_0 + \sum_{i=1}^2 \mathbf{T}_i \xi_i \rightarrow \sum_{i=0}^2 \mathbf{T}_i \xi_i \text{ with } \xi_0 = 1 \end{aligned}$$

Overall problem can be seen as:

$$\min_{x \in X} c^T x + Q(x)$$

Two-stage RO: Formulations

Min - Max - Min formulation

$$Q(x) = \max_{\xi \in \Xi} Q(x, \xi)$$

$$\min_{x \in X} c^T x + \max_{\xi \in \Xi} Q(x, \xi) = \min_{x \in X} c^T x + \max_{\xi \in \Xi} \underbrace{\min_{y \in Y} \overbrace{f(\xi)^T y}^{Q(x, \xi)}}_{\text{s.t. } W(\xi)y \leq h(\xi) - T(\xi)x}$$

- With the convention that if

$$\mathcal{Y}(x, \bar{\xi}) = \{y \in Y \mid W(\bar{\xi})y \leq h(\bar{\xi}) - T(\bar{\xi})x\} = \emptyset$$

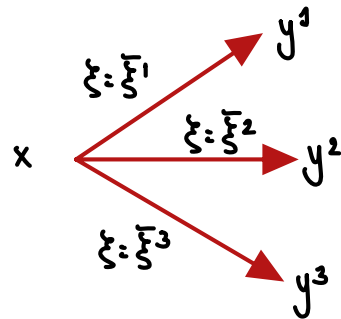
for some $\bar{\xi} \in \Xi$ then

$$\min_{y \in \mathcal{Y}(x, \bar{\xi})} f(\bar{\xi})^T y = +\infty.$$

- Useful for exact solution schemes (more on this later).

with this convention any $x \in X$
leading to infeasibility
is not a candidate for being optimal

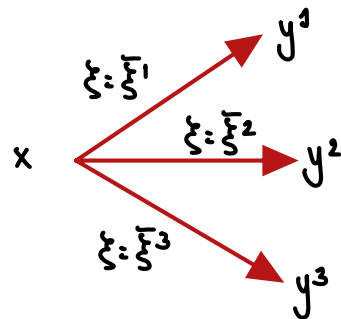
Two-stage RO: Formulations



Write :

$$\begin{aligned} \min_{x \in X} \quad & c^T x + \max_{i=1,2,3} f_i^T y_i \\ \text{s.t.} \quad & T_i x + W_i y_i \leq h_i \quad i=1,2,3 \\ & y_i \in \mathcal{Y} \quad i=1,2,3 \end{aligned}$$

$$\begin{aligned} f(\bar{\xi}^i) &= f_i & W(\bar{\xi}^i) &= W_i \\ T(\bar{\xi}^i) &= T_i & h(\bar{\xi}^i) &= h_i \end{aligned}$$



↓
If ξ is continuous then y^ξ becomes a func.
 $y(\cdot) : \Xi \rightarrow \mathcal{Y}$

$$\begin{aligned} \min_{x \in X, y(\cdot)} \quad & c^T x + \max_{\xi \in \Xi} f^T(\xi) y(\xi) \\ \text{s.t.} \quad & T(\xi) x + W(\xi) y(\xi) \leq h(\xi) \\ & y(\xi) \in \mathcal{Y} \end{aligned} \quad \left. \vphantom{\begin{aligned} \min_{x \in X, y(\cdot)} \quad & c^T x + \max_{\xi \in \Xi} f^T(\xi) y(\xi) \\ \text{s.t.} \quad & T(\xi) x + W(\xi) y(\xi) \leq h(\xi) \\ & y(\xi) \in \mathcal{Y} \end{aligned}} \right\} \forall \xi \in \Xi$$

Two-stage RO: Formulations

$$\begin{aligned}
 \min_{\mathbf{x} \in \mathcal{X}, \mathbf{y}(\cdot)} \quad & \mathbf{c}^\top \mathbf{x} + \max_{\xi \in \Xi} \mathbf{f}(\xi)^\top \mathbf{y}(\xi) \\
 \text{s.t.} \quad & \mathbf{T}(\xi)\mathbf{x} + \mathbf{W}(\xi)\mathbf{y}(\xi) \leq \mathbf{h}(\xi) & \forall \xi \in \Xi \\
 & \mathbf{y}(\xi) \in \mathcal{Y} & \forall \xi \in \Xi
 \end{aligned}$$

- $\mathbf{y}(\cdot) : \Xi \rightarrow \mathcal{Y}$ are functionals to be optimized.
- Useful for approximation schemes known as "decision rules" (more on this later).

propose a feasible sol. without an optimality certificate

Remark

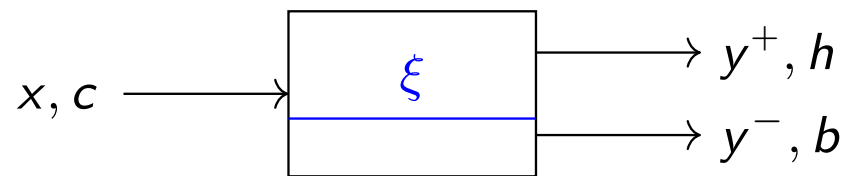
Any $\mathbf{x} \in \mathcal{X}$ for which there exists $\bar{\xi} \in \Xi$ such that

$$\mathcal{Y}(\mathbf{x}, \bar{\xi}) = \{\mathbf{y} \in \mathcal{Y} \mid \mathbf{W}(\bar{\xi})\mathbf{y} \leq \mathbf{h}(\bar{\xi}) - \mathbf{T}(\bar{\xi})\mathbf{x}\} = \emptyset$$

cannot be a feasible solution to this model.

Two-stage RO: A newsvendor example

- Order quantity x
- Uncertain demand ξ with $\xi \in \Xi = \{|\xi - \bar{\xi}| \leq \rho \bar{\xi}\}$ where $\bar{\xi} \geq 0$
- Excess quantity $y^+(\xi)$
- Shortage quantity $y^-(\xi)$
- Order cost c , return cost $h > 0$ and shortage cost $b > 0$



$$\xi \in \Xi = \{|\xi - \bar{\xi}| \leq \rho \bar{\xi}\}$$

$\hookrightarrow (1-\rho)\bar{\xi} \leq \xi \leq (1+\rho)\bar{\xi}$

Two-stage RO: A newsvendor example

- Two-stage problem written in the min-max-min form:

$$\min_{x \geq 0} \quad cx + \max_{\xi \in \Xi} \min_{y^+ \geq 0, y^- \geq 0} \quad \begin{matrix} hy^+ + by^- \\ y^+ - y^- = x - \xi \end{matrix}$$

given x & ξ
 either y^+ or $y^- > 0$
 depending on
 $x - \xi > 0$ or < 0

- Two-stage problem written in the functional form:

$$\min_{\substack{x \geq 0 \\ y^+(\cdot): \Xi \rightarrow \mathbb{R}_+ \\ y^-(\cdot): \Xi \rightarrow \mathbb{R}_+}} \quad cx + \max_{\xi \in \Xi} \quad hy^+(\xi) + by^-(\xi)$$

A feasible sol. : $y^+(\xi) = x$ $y^-(\xi) = \xi$

s.t. $y^+(\xi) - y^-(\xi) = x - \xi \quad \forall \xi \in \Xi$

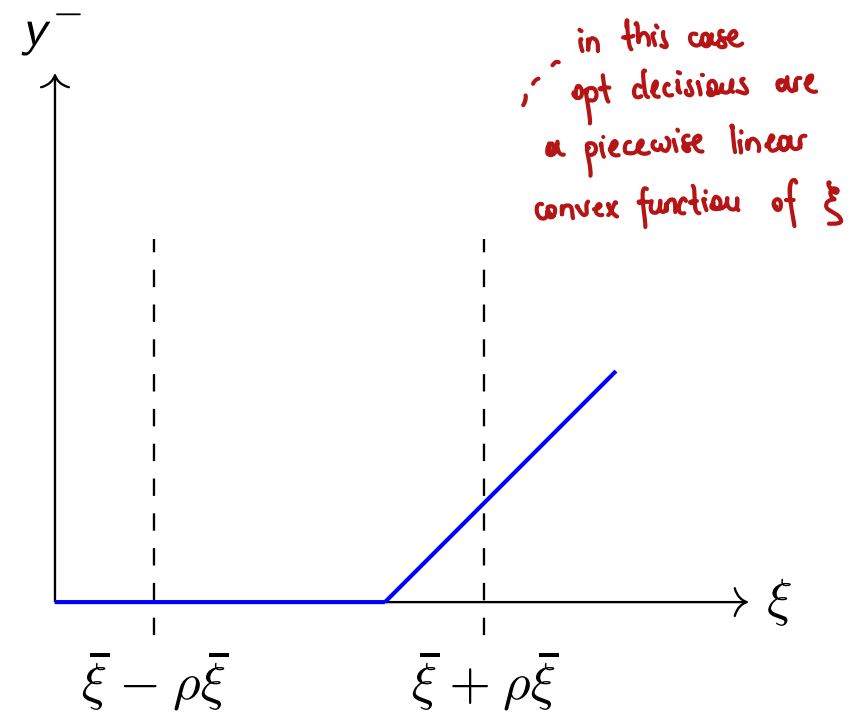
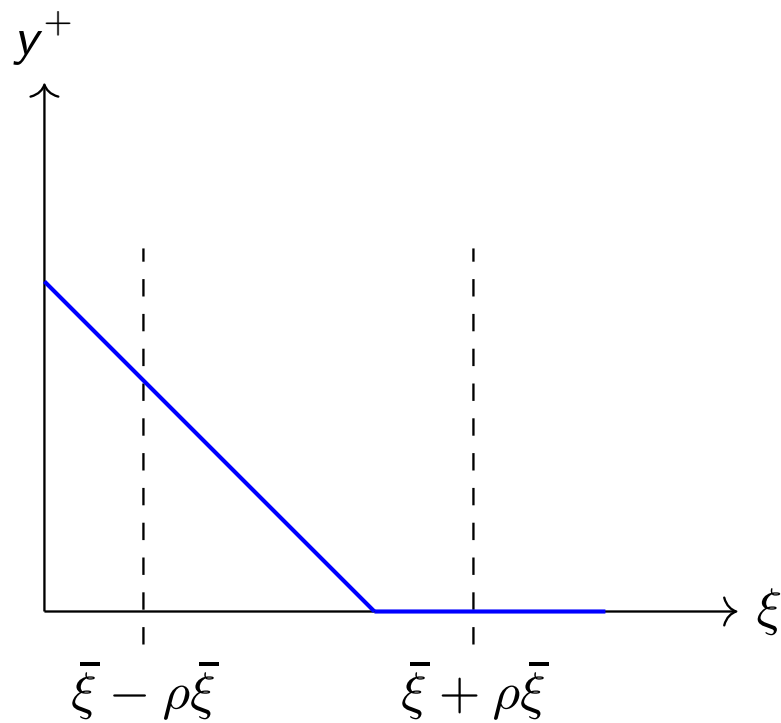
- For this problem, the optimal recourse can be written as:

$$y^+(\xi) = \max\{x - \xi, 0\}$$

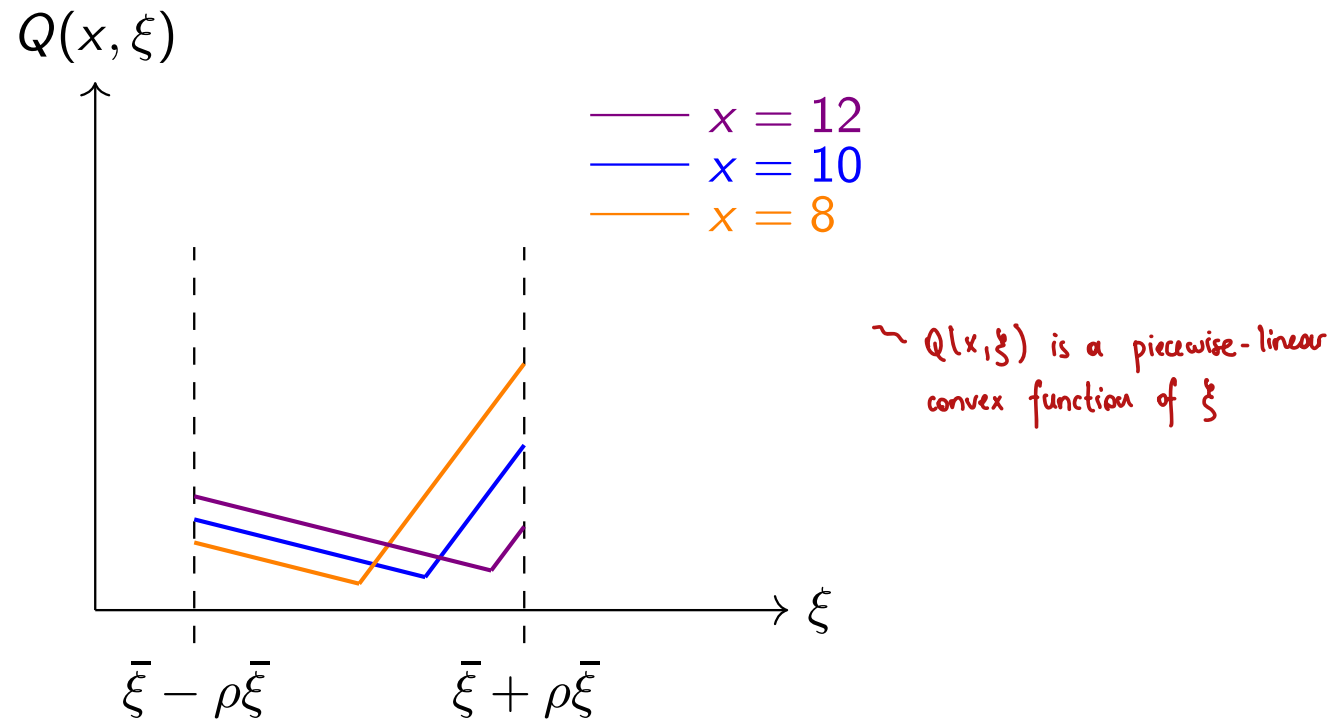
$$y^-(\xi) = \max\{0, \xi - x\}$$

Two-stage RO: A newsvendor example

- Optimal recourse quantities as a function of ξ when $x = 10$:



Two-stage RO: A newsvendor example



- When x is low worst case is $\bar{\xi} + \rho\bar{\xi}$, otherwise worst case is $\bar{\xi} - \rho\bar{\xi}$.
- Optimality is achieved for x such that:

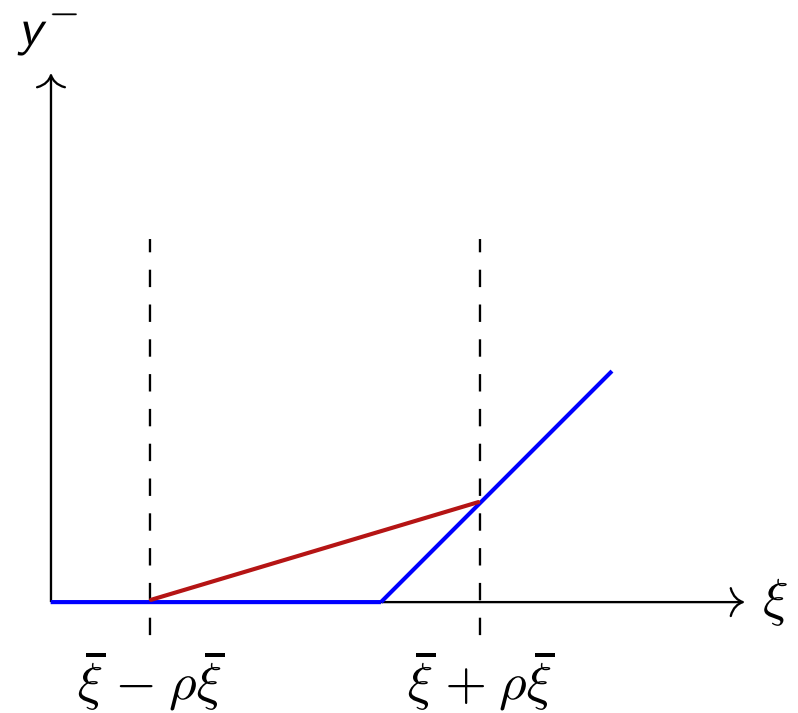
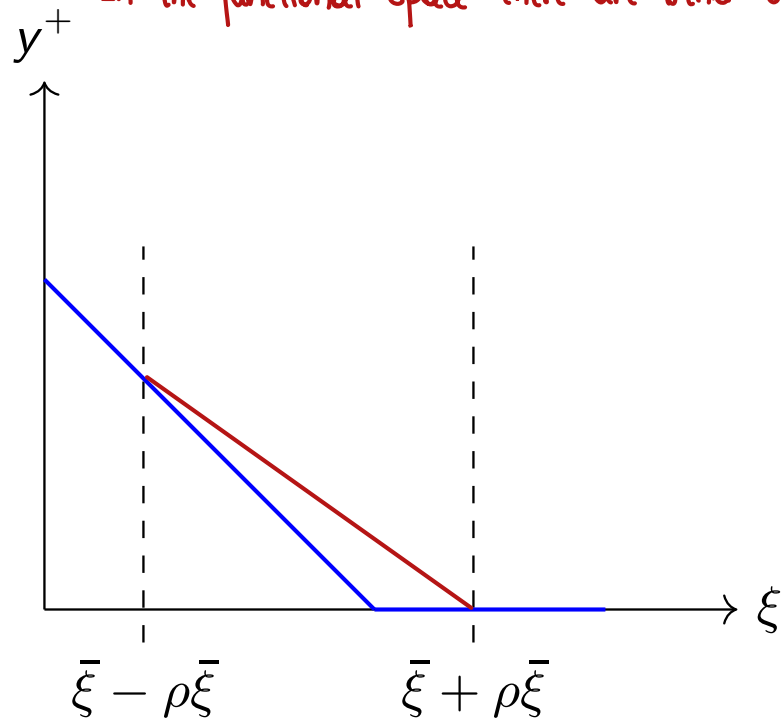
$$Q(x, \bar{\xi} + \rho\bar{\xi}) = Q(x, \bar{\xi} - \rho\bar{\xi})$$

find the balance between return & shortage costs

Two-stage RO: A newsvendor example

- Optimal recourse quantities as a function of ξ when $x = 10$:

In the functional space there are other opt. solutions



Two-stage RO: Relaxations and Restrictions

- Consider $\tilde{\Xi} \subset \Xi$:

$$\min_{\mathbf{x} \in \mathcal{X}} \quad \mathbf{c}^\top \mathbf{x} + \max_{\xi \in \tilde{\Xi}} \min_{\mathbf{y} \in \mathcal{Y}} \quad \mathbf{f}(\xi)^\top \mathbf{y}$$

s.t. $\mathbf{W}(\xi)\mathbf{y} \leq \mathbf{h}(\xi) - \mathbf{T}(\xi)\mathbf{x}$

restrict the adversary

is a relaxation of (2RO) and therefore provides a lower bound.

- In particular, for a finite subset $\tilde{\Xi} = \{\xi_1, \dots, \xi_N\} \subset \Xi$ we can write:

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}, \mathbf{y}_1, \dots, \mathbf{y}_N \in \mathcal{Y}} \quad & \mathbf{c}^\top \mathbf{x} + \theta \\ \text{s.t.} \quad & \theta \geq \mathbf{f}_\ell^\top \mathbf{y}_\ell & \ell = 1, \dots, N \\ & \mathbf{T}_\ell \mathbf{x} + \mathbf{W}_\ell \mathbf{y}_\ell \leq \mathbf{h}_\ell & \ell = 1, \dots, N \end{aligned}$$

and solve the problem as a (large-scale) (mixed-integer) linear program.

Two-stage RO: Relaxations and Restrictions

- Consider $\tilde{\mathcal{Y}} \subset \mathcal{Y}$:

$$\min_{\mathbf{x} \in \mathcal{X}} \mathbf{c}^\top \mathbf{x} + \max_{\xi \in \Xi} \min_{\substack{y \in \tilde{\mathcal{Y}} \\ \text{s.t. } \mathbf{W}(\xi)\mathbf{y} \leq \mathbf{h}(\xi) - \mathbf{T}(\xi)\mathbf{x}}} \mathbf{f}(\xi)^\top \mathbf{y}$$

restrict the recourse

is a restriction of (2RO) and therefore provides an upper bound.

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Two-stage RO with continuous and fixed recourse

$$\begin{aligned}
 \min_{\mathbf{x} \in \mathcal{X}, \mathbf{y}(\cdot)} \quad & \mathbf{c}^\top \mathbf{x} + \max_{\xi \in \Xi} \mathbf{f}(\xi)^\top \mathbf{y}(\xi) \\
 \text{s.t.} \quad & \mathbf{T}(\xi)\mathbf{x} + \mathbf{W}(\xi)\mathbf{y}(\xi) \leq \mathbf{h}(\xi) \quad \forall \xi \in \Xi \\
 & \mathbf{y}(\xi) \in \mathcal{Y} \quad \forall \xi \in \Xi
 \end{aligned}$$

Assume

- $\mathcal{Y} = \mathbb{R}_+^{n_y} \rightarrow$ continuous recourse
- $\mathbf{f}(\xi)$ and $\mathbf{W}(\xi)$ are deterministic $\rightarrow$ fixed recourse

Two-stage RO with continuous and fixed recourse

$$\min_{\mathbf{x} \in \mathcal{X}, \mathbf{y}(\cdot)} \quad \mathbf{c}^\top \mathbf{x} + \max_{\xi \in \Xi} \quad \mathbf{f}^\top \mathbf{y}(\xi)$$

$$\text{s.t.} \quad \mathbf{T}(\xi)\mathbf{x} + \mathbf{W}\mathbf{y}(\xi) \leq \mathbf{h}(\xi)$$

$$\mathbf{y}(\xi) \in \mathbb{R}_+^{n_y}$$

we'll focus on this for
the rest of the day

$$\forall \xi \in \Xi$$

$$\forall \xi \in \Xi$$

Decision rule approximations

$$\begin{aligned}
 \min_{\mathbf{x} \in \mathcal{X}, \mathbf{y}(\cdot), \theta \in \mathbb{R}} \quad & \mathbf{c}^\top \mathbf{x} + \theta \\
 \text{s.t.} \quad & \theta \geq \mathbf{f}^\top \mathbf{y}(\xi) & \forall \xi \in \Xi \\
 & \mathbf{T}(\xi)\mathbf{x} + \mathbf{W}\mathbf{y}(\xi) \leq \mathbf{h}(\xi) & \forall \xi \in \Xi \\
 & \mathbf{y}(\xi) \in \mathbb{R}_+^{n_y} & \forall \xi \in \Xi
 \end{aligned}$$

Idea

- Restrict the form of $\mathbf{y}(\xi)$ to a simple family of functions.
- Optimize within the chosen family.
- Most common restrictions: affine, piecewise affine/constant.

note: static robust optimization
can be seen as a decision rule
 $\mathbf{y}(\xi) = \mathbf{y} \quad \forall \xi \in \Xi$

Attention!

- Being restrictions, decision rule approximations may be infeasible.
- If feasible, they provide feasible solutions (primal bounds) to (2ARO).
- They typically improve upon the static robust solution. (but not always)

in the case
of min upper
bounds

Affine decision rules¹

$$\begin{aligned}
 \min_{\mathbf{x} \in \mathcal{X}, \mathbf{y}(\cdot)} \quad & \mathbf{c}^\top \mathbf{x} + \max_{\boldsymbol{\xi} \in \Xi} \mathbf{f}^\top \mathbf{y}(\boldsymbol{\xi}) \\
 \text{s.t.} \quad & \sum_{i=1}^{n_\xi} \mathbf{T}_i \xi_i \mathbf{x} + \mathbf{W} \mathbf{y}(\boldsymbol{\xi}) \leq \mathbf{H} \boldsymbol{\xi} & \forall \boldsymbol{\xi} \in \Xi \\
 & \mathbf{y}(\boldsymbol{\xi}) \geq \mathbf{0} & \forall \boldsymbol{\xi} \in \Xi
 \end{aligned}$$

Idea

- Restrict $\mathbf{y}(\boldsymbol{\xi})$ to be an affine function of $\boldsymbol{\xi}$

$$\mathbf{y}_i(\boldsymbol{\xi}) = \alpha_{i0} + \boldsymbol{\alpha}_i^\top \boldsymbol{\xi} \quad \forall i = 1, \dots, n_y \rightarrow \mathbf{y}(\boldsymbol{\xi}) = \mathbf{A} \boldsymbol{\xi} \quad \bar{\mathbf{A}} = [\boldsymbol{\alpha}_0 | \mathbf{A}] \quad \tilde{\boldsymbol{\xi}} = \begin{bmatrix} 1 \\ \boldsymbol{\xi} \end{bmatrix}$$

- Optimize $\mathbf{A} \in \mathbb{R}^{n_y \times (n_\xi + 1)}$ to obtain the best such approximation.

the parameters of the affine function become decision variables

¹Ben-Tal et al., 2004

Affine decision rules¹

$$\begin{aligned}
 & \min_{\mathbf{x} \in \mathcal{X}, \mathbf{A} \in \mathbb{R}^{n_y \times (n_\xi + 1)}, \theta \in \mathbb{R}} && \mathbf{c}^\top \mathbf{x} + \theta && (\text{Aff}) \\
 & \text{s.t.} && \theta \geq \mathbf{f}^\top \mathbf{A}\xi && \forall \xi \in \Xi \\
 & && \sum_{i=1}^{n_\xi} \mathbf{T}_i \xi_i \mathbf{x} + \mathbf{W} \mathbf{A}\xi \leq \mathbf{H}\xi && \forall \xi \in \Xi \\
 & && \mathbf{A}\xi \geq 0 && \forall \xi \in \Xi
 \end{aligned}$$

we can now see why
we need $\mathbf{f}, \mathbf{W}$
constant in ξ

Every constraint is affine
in ξ with $\mathbf{x}, \mathbf{A}, \theta$ fixed

Remark

(Aff) is a static linear robust optimization problem with a polyhedral uncertainty set $\rightarrow$ reformulate into a deterministic equivalent problem through LP-duality.

¹Ben-Tal et al., 2004

Affine decision rules¹

Take for instance: $\theta \geq f^T A \xi \quad \forall \xi \in \Xi \iff \theta \geq \max_{\xi \in \mathbb{R}^{n_\xi}} f^T A \xi$
 $D \xi \leq d \quad (u_{obj})$

$$\iff \theta \geq \min_{s.t.} d^T u_{obj}$$

$$D^T u_{obj} = f^T A$$

$$u_{obj} \geq 0$$

$$\iff \theta \geq d^T u_{obj}$$

$$D^T u_{obj} = f^T A$$

$$u_{obj} \geq 0$$

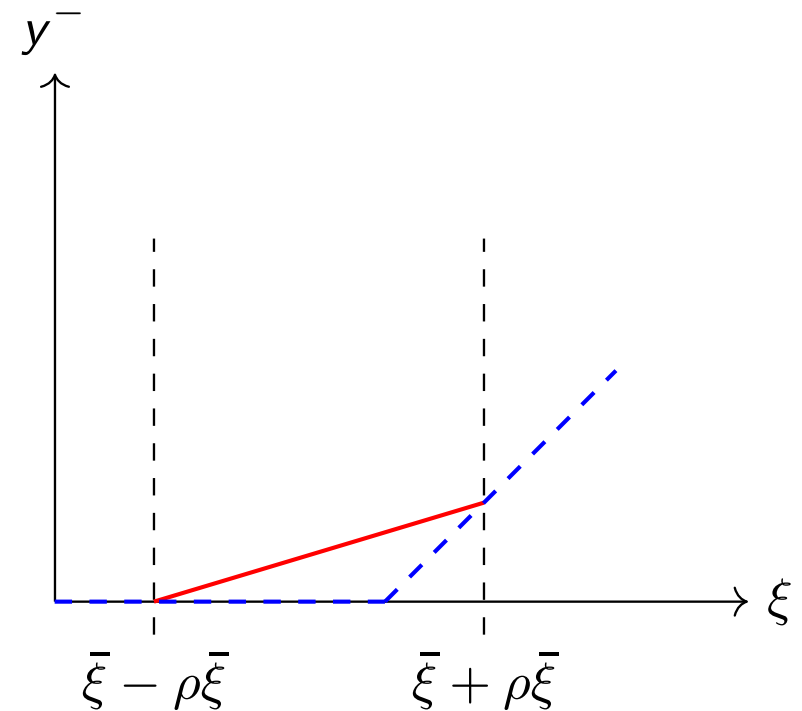
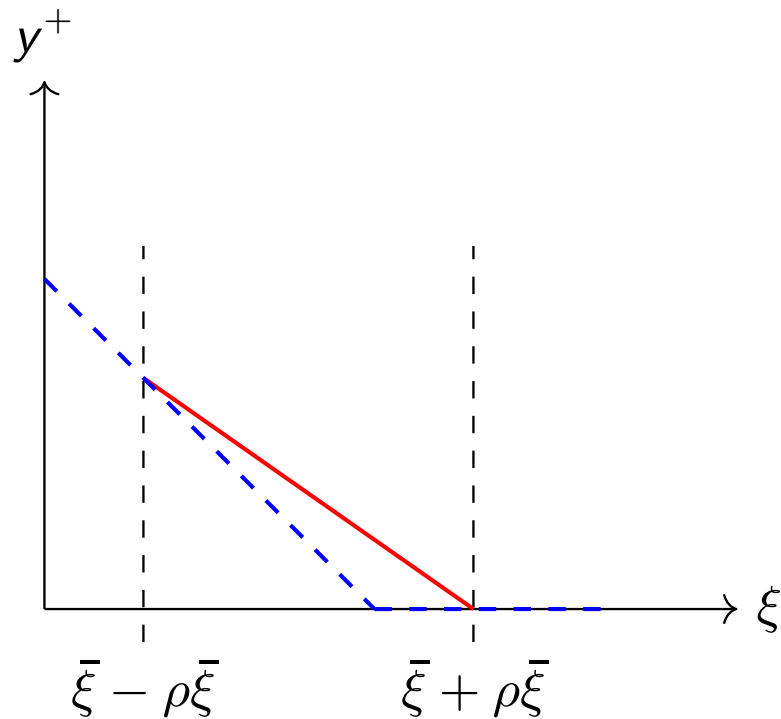
⚠ We need distinct dual variables for each semi-infinite constraint

⚠ Additional # of constraints & variables is polynomial in problem parameters

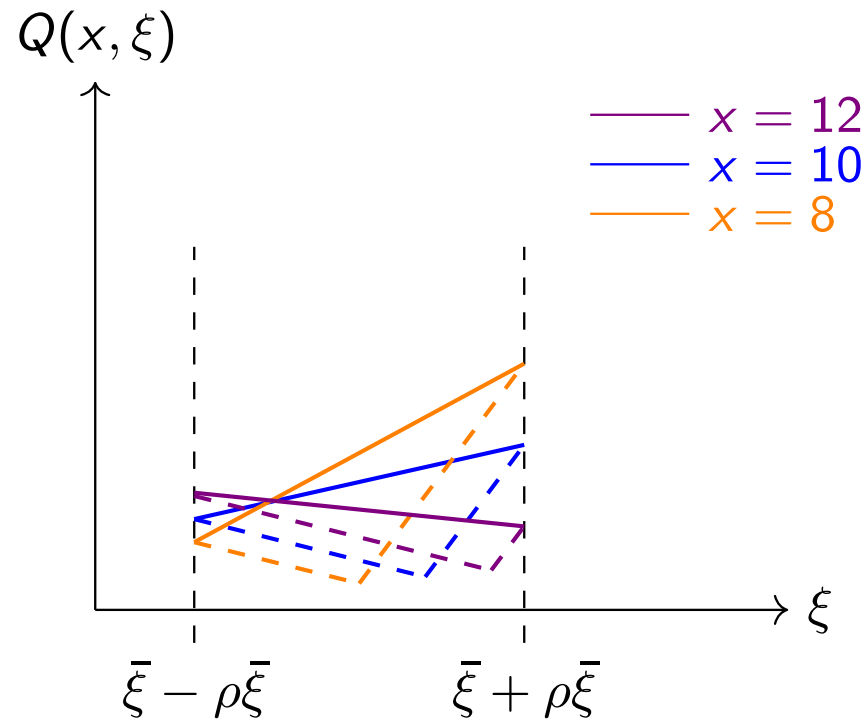
¹Ben-Tal et al., 2004

Example: Newsvendor (Cont'd)

- Let $y^+ = \alpha_0^1 + \alpha_1^1 \xi$ and $y^- = \alpha_0^2 + \alpha_1^2 \xi$
- Optimal *linear* recourse quantities as a function of ξ when $x = 10$:



Example: Newsvendor (Cont'd)



- When x is low worst case is $\bar{\xi} + \rho\bar{\xi}$, otherwise worst case is $\bar{\xi} - \rho\bar{\xi}$.
- Optimality is achieved at equality.
- Optimal x value is the same as in the exact solution².

²proved more generally in Bertsimas et al., 2010.

On the quality of affine decision rules

- The quality of a decision rule is measured based on the relative gap:

$$100 \times \frac{|z_{\text{AFF}} - z_{\text{Dual}}|}{|z_{\text{Dual}}|}$$

where z_{Dual} is a dual bound such that $z_{\text{Dual}} \leq z_{2\text{ARO}}$. *since $z_{2\text{ARO}}$ is not available*

- Let $\hat{\Xi} \subseteq \Xi$ be a finite subset of realizations.
- Then the following relaxation provides a dual bound:

$$\begin{aligned} \min_{\substack{\mathbf{x} \in \mathcal{X}, \theta \in \mathbb{R} \\ \mathbf{y}^1, \dots, \mathbf{y}^{|\hat{\Xi}|} \in \mathbb{R}_+^{n_y}}} \quad & \mathbf{c}^\top \mathbf{x} + \theta \\ \text{s.t.} \quad & \theta \geq \mathbf{f}^\top \mathbf{y}^k & \forall k \in [|\hat{\Xi}|] \\ & \mathbf{T}(\xi^k) \mathbf{x} + \mathbf{W} \mathbf{y}^k \leq \mathbf{h}(\xi^k) & \forall k \in [|\hat{\Xi}|] \end{aligned}$$

- But how do we choose $|\hat{\Xi}|$ in a meaningful way?

On the quality of affine decision rules

- The quality of a decision rule is measured based on the relative gap:

$$100 \times \frac{|z_{\text{AFF}} - z_{\text{Dual}}|}{|z_{\text{Dual}}|}$$

where z_{Dual} is a dual bound such that $z_{\text{Dual}} \leq z_{2\text{ARO}}$.

Hadjiyiannis et al. (2011)

- Solve (Aff) to optimality, let $(\mathbf{x}^*, \mathbf{A}^*, \theta^*)$ be an optimal solution.
- Extract the “binding” scenarios by solving:

$$\max_{\xi \in \Xi} \mathbf{f}^\top \mathbf{A}^* \xi - \theta^*$$

$$\max_{\xi \in \Xi} \mathbf{T}_i(\xi) \mathbf{x}^* + \mathbf{W}_i \mathbf{A}^* \xi - \mathbf{h}_i(\xi) \quad \forall i \in [m]$$

- Constitute $\hat{\Xi}$ of binding scenarios. (obtained as argmax in the above)

Monolithic reformulations (Towards exact solution algorithms)

$$\min_{\mathbf{x} \in \mathcal{X}} \mathbf{c}^\top \mathbf{x} + \max_{\xi \in \Xi} \min_{\mathbf{y} \in \mathbb{R}_+^{n_y}} \mathbf{f}^\top \mathbf{y}$$

$$\text{s.t. } \mathbf{W}\mathbf{y} \leq \mathbf{h}(\xi) - \mathbf{T}(\xi)\mathbf{x}$$

Assume

- Relatively complete recourse, i.e., $\mathcal{Y}(\mathbf{x}, \xi) \neq \emptyset$ for $\mathbf{x} \in \mathcal{X}, \xi \in \Xi$

guarantees that any solution $\mathbf{x} \in \mathcal{X}$
has value $Q(\mathbf{x}) < +\infty$

$$\min_{\mathbf{y} \in \mathcal{Y}} \mathbf{f}(\xi)^\top \mathbf{y}$$

$$\mathbf{W}(\xi)\mathbf{y} \leq \mathbf{h}(\xi) - \mathbf{T}(\xi)\mathbf{x} \quad \text{is feasible for any } \mathbf{x} \in \mathcal{X}$$

Monolithic reformulations

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}} \quad & \mathbf{c}^\top \mathbf{x} + \max_{\boldsymbol{\xi} \in \Xi} \min_{\mathbf{y} \in \mathbb{R}_+^{n_y}} \mathbf{f}^\top \mathbf{y} \\ \text{s.t.} \quad & \mathbf{W}\mathbf{y} \leq \mathbf{h}(\boldsymbol{\xi}) - \mathbf{T}(\boldsymbol{\xi})\mathbf{x} \end{aligned}$$

- Start by writing the dual of the inner minimization problem for $\mathbf{x}$ and $\boldsymbol{\xi}$ given:

$$\begin{aligned} \max_{\mathbf{u} \in \mathbb{R}_-^{m_y}} \quad & \mathbf{u}^\top (\mathbf{h}(\boldsymbol{\xi}) - \mathbf{T}(\boldsymbol{\xi})\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{W}^\top \mathbf{u} \leq \mathbf{f} \end{aligned}$$

- Since we assumed that the primal is feasible we have that the dual is bounded.
- We can also assume that the dual is feasible (why?).

Monolithic reformulations

$$Q(x, \xi) = \min_{y \in \mathbb{R}_+^{n_y}} f^\top y \quad \text{s.t.} \quad Wy \leq h(\xi) - T(\xi)x$$

strong duality

$$= \max_{u \in \mathbb{R}_-^{m_y}} u^\top (h(\xi) - T(\xi)x) \quad \text{s.t.} \quad W^\top u \leq f$$

$$D := \{ u \in \mathbb{R}_-^{m_y} \mid W^\top u \leq f \}$$

$$\max_{u \in \mathbb{R}_-^{m_y}} u^\top (h(\xi) - T(\xi)x) = \max_{u \in D} u^\top (h(\xi) - T(\xi)x) \quad \sim \text{linear programming problem}$$

linear programming problem

Let $\text{ext}(D)$ be the set of extreme points of D

$$Q(x, \xi) = \max_{u \in D} u^\top (h(\xi) - T(\xi)x) = \max_{i=1, \dots, |\text{ext}(D)|} u_i^\top (h(\xi) - T(\xi)x)$$

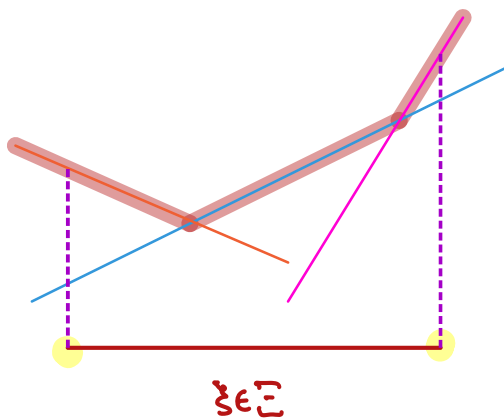
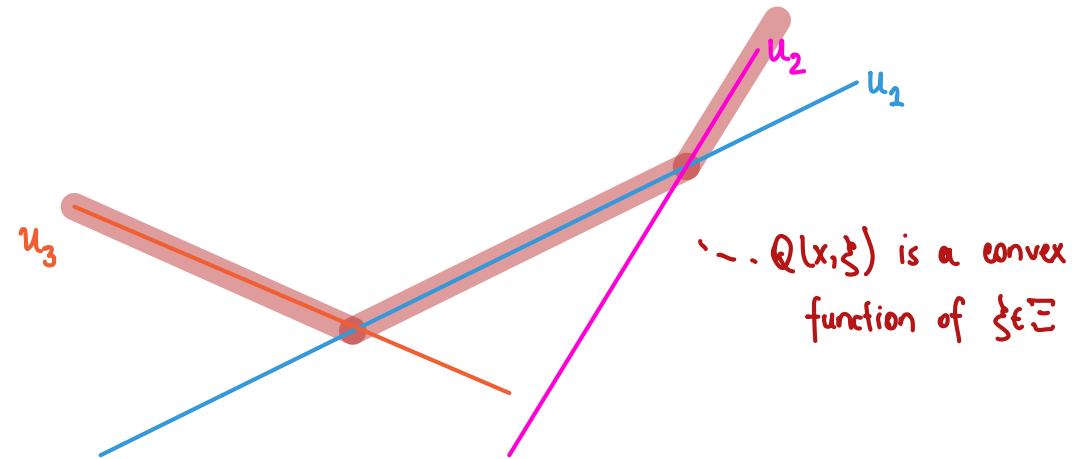
Monolithic reformulations

if $h(\xi) \in T(\xi)$ are affine functions of ξ

For x fixed $u_i^T (h(\xi) - T(\xi)x)$ defines an affine function of ξ for each $i=1, \dots, |\text{ext}(D)|$

$$Q(x, \xi) = \max_{i=1, \dots, |\text{ext}(D)|} u_i^T (h(\xi) - T(\xi)x) :$$

$$Q(x) = \max_{\xi \in \Xi} Q(x, \xi)$$



Monolithic reformulations

- Since $Q(\mathbf{x}, \xi)$ is convex in ξ , an optimal solution of $\max_{\xi \in \Xi} Q(\mathbf{x}, \xi)$ is an extreme point of Ξ .
- We may therefore write

$$\max_{\xi \in \Xi} Q(\mathbf{x}, \xi) = \max_{\xi_j \in \text{ext}(\Xi)} Q(\mathbf{x}, \xi_j) = \max_{\xi_j \in \text{ext}(\Xi), \mathbf{u}_i \in \text{ext}(\mathcal{D})} \mathbf{u}_i^\top (\mathbf{h}(\xi_j) - \mathbf{T}(\xi_j)\mathbf{x})$$

- This leads to a first exponential-sized reformulation of the problem:

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}, \theta \in \mathbb{R}} \quad & \mathbf{c}^\top \mathbf{x} + \theta \\ \text{s.t.} \quad & \theta \geq \mathbf{u}_i^\top (\mathbf{h}(\xi_j) - \mathbf{T}(\xi_j)\mathbf{x}) \quad \forall \xi_j \in \text{ext}(\Xi), \mathbf{u}_i \in \text{ext}(\mathcal{D}) \end{aligned}$$

- monolithic & linear
- exponential sized
- need to enumerate $\text{ext}(\Xi), \text{ext}(\mathcal{D})$

Monolithic reformulations

- For each $\xi_j \in \text{ext}(\Xi)$, there must be a ^{optimal} recourse solution $\mathbf{y}_j^* \in \mathbb{R}_+^{n_y}$ such that

$$\mathbf{W}\mathbf{y}_j^* \leq \mathbf{h}(\xi) - \mathbf{T}(\xi)\mathbf{x}$$

- We then have that

$$\max_{\xi \in \Xi} Q(\mathbf{x}, \xi) = \max_{\xi_j \in \text{ext}(\Xi)} Q(\mathbf{x}, \xi_j) = \max_{j=1, \dots, |\text{ext}(\Xi)|} \mathbf{f}^\top \mathbf{y}_j^*$$

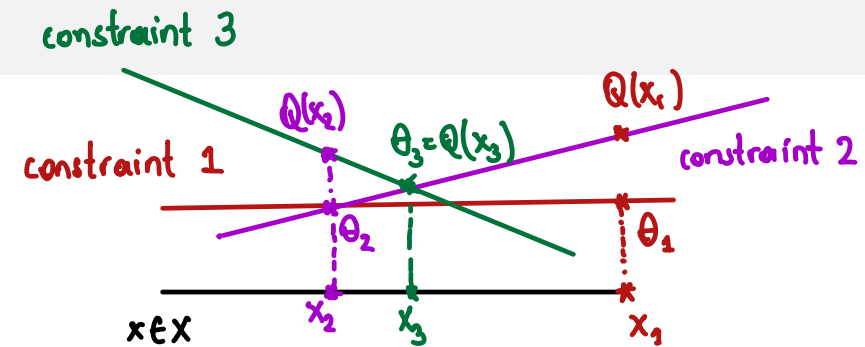
- Which leads to yet another exponential size formulation of the problem:

$$\begin{aligned} \min_{\substack{\mathbf{x} \in \mathcal{X}, \theta \in \mathbb{R} \\ \mathbf{y}_1, \dots, \mathbf{y}_{|\text{ext}(\Xi)|} \in \mathbb{R}_+^{n_y}}} \quad & \mathbf{c}^\top \mathbf{x} + \theta \\ \text{s.t.} \quad & \theta \geq \mathbf{f}^\top \mathbf{y}_j & j = 1, \dots, |\text{ext}(\Xi)| \\ & \mathbf{T}(\xi_j)\mathbf{x} + \mathbf{W}\mathbf{y}_j \leq \mathbf{h}(\xi_j) & j = 1, \dots, |\text{ext}(\Xi)| \end{aligned}$$

y_j are decision vars & will take the necessary values to push θ down

*- monolithic & linear
- exponential sized
- need to enumerate ext(Ξ)*

Decomposition algorithms



$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & \mathbf{c}^\top \mathbf{x} + \theta \\ \text{s.t.} \quad & \theta \geq \dots \end{aligned}$$

Idea (Very high level)

- Solve a relaxation of the problem by including only a subset of the constraints on θ .
- Let $(\mathbf{x}^*, \theta^*)$ be an optimal relaxation solution.
- Solve $\max_{\xi \in \Xi} Q(\mathbf{x}^*, \xi)$ to calculate the worst-case value of $\mathbf{x}^*$.
- If $\theta^* < \max_{\xi \in \Xi} Q(\mathbf{x}^*, \xi)$ then add constraints on θ .
- Otherwise $(\mathbf{x}^*, \theta^*)$ is an optimal solution.

all necessary constraints are identified

Aim to iteratively reconstruct $Q(x)$ around an opt. sol.

Constraint-generation algorithm³

$$\max_{\mathbf{x} \in \Xi} Q(\mathbf{x}, \xi) = \max_{\xi_j \in \text{ext}(\Xi)} Q(\mathbf{x}, \xi_j) = \max_{\xi_j \in \text{ext}(\Xi), \mathbf{u}_i \in \text{ext}(\mathcal{D})} \mathbf{u}_i^\top (\mathbf{h}(\xi_j) - \mathbf{T}(\xi_j)\mathbf{x})$$

- Solve the relaxation:

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}, \theta \in \mathbb{R}} \quad & \mathbf{c}^\top \mathbf{x} + \theta \\ \text{s.t.} \quad & \theta \geq \mathbf{u}_\ell^\top (\mathbf{h}(\xi_\ell) - \mathbf{T}(\xi_\ell)\mathbf{x}) \quad \ell = 1, \dots, N \end{aligned}$$

- Let $(\mathbf{x}^*, \theta^*)$ be an optimal solution.
- Solve $\max_{\xi \in \Xi} Q(\mathbf{x}^*, \xi)$. *must exist since otherwise $\theta^* = Q(\mathbf{x}^*)$*
- If $\theta^* < Q(\mathbf{x}^*)$, let $(\mathbf{u}_{N+1}, \xi_{N+1})$ define a violated constraint.
- Add the constraint:

$$\theta \geq \mathbf{u}_{N+1}^\top (\mathbf{h}(\xi_{N+1}) - \mathbf{T}(\xi_{N+1})\mathbf{x})$$

to refine the relaxation.

- Otherwise $(\mathbf{x}^*, \theta^*)$ is an optimal solution.

*at most $|\text{ext}(\Xi)| \times |\text{ext}(\mathcal{D})|$ iterations
(assuming $(\mathbf{u}, \xi)$ are extreme points of $\mathcal{D} \times \Xi$)*

³Thiele, 2009, Bertsimas et al., 2013

Constraint-and-column generation (CCG) algorithm⁴

$$\max_{\xi \in \Xi} Q(\mathbf{x}, \xi) = \max_{\xi_j \in \text{ext}(\Xi)} Q(\mathbf{x}, \xi_j) = \max_{j=1, \dots, |\text{ext}(\Xi)|} \mathbf{f}^\top \mathbf{y}_j^*$$

- Consider a discrete set $\tilde{\Xi} = \{\xi_1, \dots, \xi_N\} \subset \text{ext}(\Xi)$, and solve the relaxation:

$$\begin{aligned} \min_{\substack{\mathbf{x} \in \mathcal{X}, \theta \in \mathbb{R} \\ \mathbf{y}_1, \dots, \mathbf{y}_N \in \mathbb{R}_+^{n_y}}} \quad & \mathbf{c}^\top \mathbf{x} + \theta \\ \text{s.t.} \quad & \theta \geq \mathbf{f}^\top \mathbf{y}_\ell & \ell = 1, \dots, N \\ & \mathbf{T}(\xi_\ell) \mathbf{x} + \mathbf{W} \mathbf{y}_\ell \leq \mathbf{h}(\xi_\ell) & \ell = 1, \dots, N. \end{aligned}$$

- Let $(\mathbf{x}^*, \theta^*)$ be an optimal solution.
- Solve $\max_{\xi \in \Xi} Q(\mathbf{x}^*, \xi)$. *must exist since otherwise $\theta^* = Q(\mathbf{x}^*)$*
- If $\theta^* < Q(\mathbf{x}^*)$, let ξ_{N+1} be a realization that needs to be added to $\tilde{\Xi}$.
- Add variables $\mathbf{y}_{N+1} \in \mathbb{R}_+^{n_y}$ and constraints:

$$\begin{aligned} \theta &\geq \mathbf{f}^\top \mathbf{y}_{N+1} \\ \mathbf{T}(\xi_{N+1}) \mathbf{x} + \mathbf{W} \mathbf{y}_{N+1} &\leq \mathbf{h}(\xi_{N+1}) \end{aligned}$$

*at most $|\Xi|$ iterations
(assuming ξ are
extreme points of Ξ)*

- Otherwise $(\mathbf{x}^*, \theta^*)$ is an optimal solution.

⁴Zeng and Zhao, 2013

$$\max_{\xi \in \Xi} Q(x, \xi) :$$

- At each iteration of decomposition algorithms, we need to solve:

$$\begin{aligned} \max_{\xi \in \Xi} \quad & Q(\mathbf{x}^*, \xi) = \max_{\xi \in \Xi} \min_{\mathbf{y} \in \mathbb{R}_+^{n_y}} \quad \mathbf{f}^\top \mathbf{y} \\ & \text{convex func. of } \xi \\ & \text{not explicitly known} \\ \text{s.t.} \quad & \mathbf{W}\mathbf{y} \leq \mathbf{h}(\xi) - \mathbf{T}(\xi)\mathbf{x}^* \end{aligned}$$

to evaluate the true worst-case value of a given first-stage solution $\mathbf{x}^*$.

- This is a difficult problem since it amounts to maximizing a convex function.
- Here we will talk about generic exact approaches based on reformulation as a MIP.

Separation problem

- Write the dual of the inner minimization problem for $\mathbf{x}^*$ and ξ given:

$$\begin{aligned} Q(\mathbf{x}^*, \xi) = \max_{\mathbf{u} \in \mathbb{R}_-^{m_y}} \quad & \mathbf{u}^\top (\mathbf{h}(\xi) - \mathbf{T}(\xi)\mathbf{x}^*) \\ \text{s.t.} \quad & \mathbf{W}^\top \mathbf{u} \leq \mathbf{f} \end{aligned}$$

- Merge the two max problems to obtain the **bilinear** subproblem:

disjoint bilinear $\dashrightarrow$

$$\begin{aligned} \max_{\xi^* \in \text{ext}(\Xi), \mathbf{u}^* \in \text{ext}(\mathcal{D})} \quad & -\mathbf{u}^\top (\mathbf{h}(\xi)) - \mathbf{T}(\xi)\mathbf{x}^* \\ \text{s.t.} \quad & -\mathbf{W}^\top \mathbf{u} \leq \mathbf{f} \\ & \mathbf{u} \in \mathbb{R}_+^{m_y} \\ & \xi \in \Xi := \{\xi \in \mathbb{R}_+^{n_\xi} : \mathbf{D}\xi \leq \mathbf{d}\}. \end{aligned}$$

$\mathbf{u}^\top \mathbf{h}(\xi) = \mathbf{u}^\top \mathbf{H} \xi = \sum_{i,j} h_{ij} u_i \xi_j$
bilinear

- Can be linearized if $\xi \in \{0, 1\}^{n_\xi}$ for any extreme point solution of Ξ .

Separation problem

- Assume that $\xi_j \in \{0, 1\}$ for $j = 1, \dots, n_\xi$ and $0 \leq u_i \leq M_i$ for $i = 1, \dots, m_y$.
- Replace each bilinear term $u_i \times \xi_j$ with the auxiliary variable ζ_{ij} . $\rightarrow \zeta_{ij} = u_i \times \xi_j$
- Introduce the linearization constraints (McCormick envelope):

$$\left. \begin{aligned} \zeta_{ij} &\leq u_i \\ \zeta_{ij} &\leq M_i \xi_j \\ \zeta_{ij} &\geq u_i + M_i(\xi_j - 1) \\ \zeta_{ij} &\geq 0 \end{aligned} \right\}$$

$$\begin{aligned} \xi_j = 1 &\rightarrow \zeta_{ij} = u_i \\ \xi_j = 0 &\rightarrow \zeta_{ij} = 0 \end{aligned}$$

exists since dual pb. is bounded

Remark

The extreme points of the budgeted uncertainty set

$$\Xi^\Gamma = \left\{ \xi \in [0, 1]^{n_\xi} \left| \sum_{i=1}^{n_\xi} \xi_i \leq \Gamma \right. \right\}$$

are binary vectors, $\xi \in \{0, 1\}^{n_\xi}$, when Γ is integer.

$$\sum_{i,j} h_{ij} u_i \xi_j$$

$$\rightarrow \sum_{i,j} h_{ij} \zeta_{ij}$$

$$\zeta_{ij} \leq u_i$$

$$\zeta_{ij} \leq M_i \xi_j$$

$$\zeta_{ij} \geq u_i + M_i(\xi_j - 1)$$

$$\zeta_{ij} \geq 0$$

Separation problem

① When $\Gamma \notin \mathbb{Z}$, we can transform Ξ^Γ as follows:

$$\tilde{\Xi}^\Gamma = \left\{ \xi^1, \xi^2 \in [0,1]^{n_\xi} \mid \sum_{i=1}^{n_\xi} \xi_i^1 \leq \lfloor \Gamma \rfloor, \sum_{i=1}^{n_\xi} \xi_i^2 \leq 1, \xi_i^1 + \xi_i^2 \leq 1 \quad \forall i \in [n_\xi] \right\}$$

data is transformed as: $\xi_i \rightarrow \xi_i^1 + (\Gamma - \lfloor \Gamma \rfloor) \xi_i^2$

② More generally: assume that we can write $\xi = Aw$ with $w \in \Omega \subseteq \mathbb{R}^{n_w}$ s.t.
extreme points of Ω are $\{0,1\}$.

$u^\top \xi \rightarrow u^\top Aw$: $u_i w_j$ is linearized using $w_j \in \{0,1\}$

Separation problem *(when extreme points of Ξ are not $\in \{0,1\}^{\xi}$)*

- Use the LP duality conditions on the inner problem to write:

$$\begin{aligned}
 & \max \quad \mathbf{f}^\top \mathbf{y} \\
 & \text{s.t.} \quad \mathbf{W}\mathbf{y} \leq \mathbf{h}(\boldsymbol{\xi}) - \mathbf{T}(\boldsymbol{\xi})\mathbf{x}^* \quad \left. \begin{array}{l} \\ \mathbf{y} \in \mathbb{R}_+^{n_y} \end{array} \right\} \text{Primal feasibility} \\
 & \quad \quad -\mathbf{W}^\top \mathbf{u} \leq \mathbf{f} \quad \left. \begin{array}{l} \\ \mathbf{u} \in \mathbb{R}_+^{m_y} \end{array} \right\} \text{Dual feasibility} \\
 & \text{CS} \quad \left\{ \begin{array}{ll} \mathbf{u}_i \times (\mathbf{h}(\boldsymbol{\xi}) - \mathbf{T}(\boldsymbol{\xi})\mathbf{x}^* - \mathbf{W}\mathbf{y})_i = 0 & \forall i = 1, \dots, m_y \\ (\mathbf{W}^\top \mathbf{u} + \mathbf{f})_j \times \mathbf{y}_j = 0 & \forall j = 1, \dots, n_y \end{array} \right. \\
 & \quad \boldsymbol{\xi} \in \Xi := \{\boldsymbol{\xi} \in \mathbb{R}_+^{n_\xi} : \mathbf{D}\boldsymbol{\xi} \leq \mathbf{d}\}
 \end{aligned}$$

- To handle the bilinear CS constraints, linearize with big-M constraints.
- Introduce an auxiliary variable $\zeta_i \in \{0, 1\}$ for $i = 1, \dots, m_y$ and write:

$$\begin{aligned}
 \zeta_i = 0 & \rightarrow u_i = 0 & \mathbf{u}_i = 0 & \vee & (\mathbf{h}(\boldsymbol{\xi}) - \mathbf{T}(\boldsymbol{\xi})\mathbf{x}^* - \mathbf{W}\mathbf{y})_i = 0 \\
 \zeta_i = 1 & \rightarrow (\mathbf{W}\mathbf{y} - \mathbf{h}(\boldsymbol{\xi}) + \mathbf{T}(\boldsymbol{\xi})\mathbf{x}^*)_i = 0 & & \iff & \\
 & & \mathbf{u}_i \leq M\zeta_i & & (\mathbf{h}(\boldsymbol{\xi}) - \mathbf{T}(\boldsymbol{\xi})\mathbf{x}^* - \mathbf{W}\mathbf{y})_i \leq M(1 - \zeta_i)
 \end{aligned}$$

Handling infeasibility in recourse

- Up to now we worked with a relatively complete recourse assumption.
- When this assumption is not satisfied it is possible for $\mathcal{Y}(\mathbf{x}, \xi)$ to be empty for some $\mathbf{x} \in \mathcal{X}$ and $\xi \in \Xi$.
- Assume now that $\mathcal{Y}(\bar{\mathbf{x}}, \bar{\xi}) = \emptyset$ for some $\bar{\mathbf{x}} \in \mathcal{X}$ and $\bar{\xi} \in \Xi$.
- Then

$$\begin{aligned} \max_{\mathbf{u} \in \mathbb{R}_+^{m_y}} \quad & \mathbf{u}^\top (\mathbf{h}(\bar{\xi}) - \mathbf{T}(\bar{\xi})\bar{\mathbf{x}}) \\ \text{s.t.} \quad & \mathbf{W}^\top \mathbf{u} \leq \mathbf{f} \end{aligned}$$

is unbounded. (why?)

How to cut off
infeasible $\bar{\mathbf{x}} \in \mathcal{X}$?

Handling infeasibility in recourse

$$\begin{aligned} \max_{\mathbf{u} \in \mathbb{R}_-^{m_y}} \quad & \mathbf{u}^\top (\mathbf{h}(\bar{\xi}) - \mathbf{T}(\bar{\xi})\bar{\mathbf{x}}) \\ \text{s.t.} \quad & \mathbf{W}^\top \mathbf{u} \leq \mathbf{f} \end{aligned}$$

- For C&CG, simply add variable $\mathbf{y}_{\bar{\xi}} \in \mathbb{R}_+^{n_y}$ and constraints

$$\theta \geq \mathbf{f}^\top \mathbf{y}_{\bar{\xi}}$$

$$\mathbf{T}(\bar{\xi})\bar{\mathbf{x}} + \mathbf{W}\mathbf{y}_{\bar{\xi}} \leq \mathbf{h}(\bar{\xi})$$

- For CG note that dual unbounded implies existence

of $\mathbf{u} \in \mathbb{R}_-^{m_y}$ s.t. starting from $\bar{\mathbf{u}} \in \mathbb{R}_-^{m_y}$, $\mathbf{W}^\top \bar{\mathbf{u}} \leq \mathbf{f}$

$\bar{\mathbf{u}} + \lambda \mathbf{u}$ is feasible $\forall \lambda > 0$ &

$$\mathbf{u}^\top (\mathbf{h}(\bar{\xi}) - \mathbf{T}(\bar{\xi})\bar{\mathbf{x}}) > 0$$

Add the constraint: $\mathbf{u}^\top (\mathbf{h}(\bar{\xi}) - \mathbf{T}(\bar{\xi})\bar{\mathbf{x}}) \leq 0$

Handling infeasibility in recourse

- MIP reformulations of the max-min problem require the inner minimization problem to be feasible (its dual to be bounded).
- How do we solve

$$\begin{aligned} \max_{\xi \in \Xi} Q(\mathbf{x}^*, \xi) = \max_{\xi \in \Xi} \min_{\mathbf{y} \in \mathbb{R}_+^{n_y}} \quad & \mathbf{f}^\top \mathbf{y} \\ \text{s.t.} \quad & \mathbf{W}\mathbf{y} \leq \mathbf{h}(\xi) - \mathbf{T}(\xi)\mathbf{x}^* \end{aligned}$$

when the inner min is infeasible (its dual is unbounded)?

Handling infeasibility in recourse

- Consider the recourse problem written as a feasibility problem for given $(\bar{\mathbf{x}}, \bar{\theta})$:

$$\begin{aligned}
 \min_{\mathbf{y} \in \mathbb{R}_+^{n_y}} \quad & 0 \\
 \text{s.t.} \quad & \mathbf{f}^\top \mathbf{y} \leq \bar{\theta} \\
 & \mathbf{W}\mathbf{y} \leq \mathbf{h}(\bar{\xi}) - \mathbf{T}(\bar{\xi})\bar{\mathbf{x}}
 \end{aligned}$$

① no feasible $\mathbf{y}$ s.t. $\mathbf{f}^\top \mathbf{y} \leq \bar{\theta}$
 ② $\mathbf{W}\mathbf{y} \leq \mathbf{h}(\bar{\xi}) - \mathbf{T}(\bar{\xi})\bar{\mathbf{x}}$ cannot be satisfied
 $(\sigma \in \mathbb{R}_-)$
 $(\mathbf{u} \in \mathbb{R}_-^{m_y})$

- Its dual is then given as:

$$\begin{aligned}
 \max_{\mathbf{u} \in \mathbb{R}_-^{m_y}, \sigma \in \mathbb{R}_-} \quad & \sigma \bar{\theta} + \mathbf{u}^\top (\mathbf{h}(\bar{\xi}) - \mathbf{T}(\bar{\xi})\bar{\mathbf{x}}) \\
 \text{s.t.} \quad & \mathbf{f}\sigma + \mathbf{W}\mathbf{u} \leq 0
 \end{aligned}$$

$(\sigma, \mathbf{u}) = (0, 0)$
 is a feasible solution

- The optimal value of the dual problem is greater than or equal to 0.
- The optimal value is equal to 0 only if the primal problem is feasible.

Handling infeasibility in recourse

- Further, any feasible solution $(\sigma, \mathbf{u})$ can be scaled to obtain another feasible solution.
- The dual variables can be bounded without changing the conclusion:

$$\begin{aligned}
 & \max_{\mathbf{u} \in \mathbb{R}_-^{m_y}, \sigma \in \mathbb{R}_-} && \sigma \bar{\theta} + \mathbf{u}^\top (\mathbf{h}(\bar{\xi}) - \mathbf{T}(\bar{\xi})\bar{\mathbf{x}}) \\
 & \text{s.t.} && \mathbf{f}\sigma + \mathbf{W}\mathbf{u} \leq 0 \\
 & && |\sigma| + \sum_{i=1}^{m_y} |u_i| \leq 1
 \end{aligned}$$

*either obj > 0 → dual unbd.
primal infeas.*

OR obj = 0 → primal/dual feas.

Handling infeasibility in recourse

- We can then write the separation problem:

$$\begin{aligned}
 & \max_{\xi \in \Xi, \mathbf{u} \in \mathbb{R}_-^{m_y}, \sigma \in \mathbb{R}_-} \quad \sigma \bar{\theta} + \mathbf{u}^\top (\mathbf{h}(\xi) - \mathbf{T}(\xi) \bar{\mathbf{x}}) \\
 & \text{s.t.} \quad \mathbf{f}\sigma + \mathbf{W}\mathbf{u} \leq 0 \\
 & \quad \quad |\sigma| + \sum_{i=1}^{m_y} |u_i| \leq 1
 \end{aligned}$$

- Bilinear terms can be linearized if $\xi \in \{0, 1\}^{n_\xi}$ for any extreme point solution of Ξ .
- Dual variables can be bounded by 1 when necessary.
- If $\xi \notin \{0, 1\}^{n_\xi}$ for any extreme point solution of Ξ then we use **KKT** conditions after introducing artificial variables.

LP-optimality

Handling infeasibility in recourse

- When the optimal value of the separation problem is > 0 then we need to cut off the current solution $\bar{\mathbf{x}} \in \mathcal{X}$.

- For the constraint generation algorithm, we add:

$$\sigma^* \bar{\theta} + \mathbf{u}^{*\top} (\mathbf{h}(\xi^*) - \mathbf{T}(\xi^*) \bar{\mathbf{x}}) \leq 0$$

Two cases:

① $\sigma = 0 \rightarrow \mathbf{u}^\top (\mathbf{h}(\xi^*) - \mathbf{T}(\xi^*) \bar{\mathbf{x}}) \leq 0$

② $\sigma < 0 \rightarrow$
 $\theta \geq \bar{\mathbf{u}}^{*\top} (\mathbf{h}(\xi^*) - \mathbf{T}(\xi^*) \bar{\mathbf{x}})$

with $(\xi^*, \mathbf{u}^*, \sigma^*)$ being an optimal solution of the separation problem.

- For the constraint-and-column generation algorithm, we add variables $\mathbf{y}_{\xi^*} \in \mathbb{R}_+^{n_y}$ and constraints:

$$\theta \geq \mathbf{f}^\top \mathbf{y}_{\xi^*}$$

$$\mathbf{T}(\xi^*) \mathbf{x} + \mathbf{W} \mathbf{y}_{\xi^*} \leq \mathbf{h}(\xi^*)$$

Appendix

Two-stage RO: Why the sup?

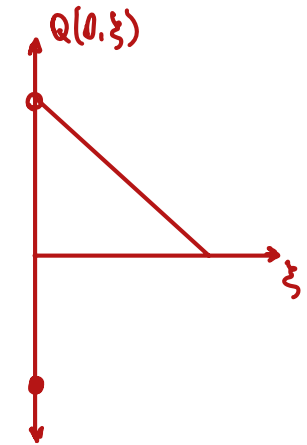
$$x=0, \xi=0 \longrightarrow \min_{y \in \{0,1\} : y \geq 0} 2y - 1 = -1$$

$$x=0, \xi > 0 \longrightarrow \min_{y \in \{0,1\} : y \geq \xi} (\xi - 1)(1 - 2y) = 1 - \xi$$

- Consider¹:

$$\min_{x \in \{0,1\}} \sup_{\xi \in [0,1]} \min_{y \in \{0,1\} : y \geq \xi, y \geq x} (\xi - 1)(1 - 2y)$$

- Fix $x = 0$.
- If $\xi = 0$ then $y = 0$ is optimal with value -1 .
- If $\xi > 0$ then $y = 1$ is optimal with value $1 - \xi$.
- The function $Q(x=0, \xi)$ is discontinuous at $\xi = 0$.



$\hookrightarrow \max_{\xi \in \Xi} Q(x, \xi)$ does not have an optimal solution

Remark

In the following, we will assume that $Q(x, \xi)$ is an upper semi-continuous function in $\xi \in \Xi$ and replace the sup with a max.

1. Continuous recourse and $f(\xi) = f$, $W(\xi) = W$ for all $\xi \in \Xi$

2. Mixed-integer recourse and only the objective function is uncertain. ($f(\xi) = f$, $W(\xi) = W$, $T(\xi) = T$ for all $\xi \in \Xi$)

¹adapted from Subramanyam et al. (19)

$$\min_{x \in \mathcal{X}} c^T x + \max_{\xi \in \Xi} Q(x, \xi)$$

$$\begin{aligned} \min_{x \in \mathcal{X}, y(\cdot)} \quad & c^T x + \max_{\xi \in \Xi} f(\xi)^T y(\xi) \\ \text{s.t.} \quad & T(\xi)x + W(\xi)y(\xi) \leq h(\xi) \\ & y(\xi) \in \mathcal{Y} \end{aligned}$$

$$\forall \xi \in \Xi$$

$$\forall \xi \in \Xi$$

$$\begin{aligned} Q(x, \xi) = \min_y \quad & f(\xi)^T y \\ \text{s.t.} \quad & W(\xi)y \leq h(\xi) - T(\xi)x \\ & y \in \mathcal{Y} \end{aligned}$$

are these two models equivalent?

① Feasibility:

In min-max-min if $\exists x \in X$ s.t. $\{y \in \mathcal{Y} \mid W(\xi)y \leq h(\xi) - T(\xi)x\} \neq \emptyset \quad \forall \xi \in \Xi$

then $x \in X$ s.t. $\exists \bar{\xi} \in \Xi$ s.t. $\{y \in \mathcal{Y} \mid W(\bar{\xi})y \leq h(\bar{\xi}) - T(\bar{\xi})x\} = \emptyset$ cannot be optimal.

In functional for $x \in X$ to be feasible, we need

$$\exists y \in \mathcal{Y} \quad \text{for } \forall \xi \in \Xi \quad \text{s.t. } T(\xi)x + W(\xi)y(\xi) \leq h(\xi)$$

↳ Set of $x \in X$ s.t. $Q(x) < +\infty$ in min-max-min & set of feasible $x \in X$ in functional are the same.

② Optimality:

Take an opt. sol of functional $(x, y(\cdot))$ value = $c^T x + \max_{\xi \in \Xi} f^T(\xi)y(\xi)$

If we take x part of this sol. in max-min do we have

$$c^T x + \max_{\xi \in \Xi} Q(x, \xi) = c^T x + \max_{\xi \in \Xi} f^T(\xi)y(\xi) ?$$

Suppose not then $\max_{\xi \in \Xi} Q(x, \xi) < \max_{\xi \in \Xi} f^T(\xi)y(\xi)$

implies $\forall \xi \in \Xi \quad Q(x, \xi) < \max_{\xi \in \Xi} f^T(\xi)y(\xi)$ & for $\xi' \in \operatorname{argmax}_{\xi \in \Xi} f^T(\xi)y(\xi)$ we must have

that $Q(x, \xi') < f^T(\xi')y(\xi')$

By replacing $y(\xi')$ with $y' \in \operatorname{argmin}_{y \in \mathcal{Y}(x, \xi')} Q(x, \xi')$ we still have $y(\cdot)$ feasible

and with smaller value.

X contradiction to optimality of $y(\cdot)$.

Similarly we can obtain an optimal policy $(x, y(\cdot))$ from an opt. solution x of min-max-min with the same value.

Note however that for the same x , there may exist many $y(\cdot)$ that are optimal but

not necessarily $y(\xi) \in \operatorname{argmin}_{y \in \gamma(x, \xi)} f(\xi)^T y$

